

When More Epochs Stop Helping: Coverage-Limited Scaling Laws for Sparse Replay

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Abstract

Repeated passes over a fixed dataset can improve optimization, but they cannot reveal predictive features absent from the unique examples. We formalize this distinction in sparse power-law regression, where feature j activates with probability $p_j \asymp j^{-s}$, has covariance eigenvalue $\lambda_j \asymp j^{-a}$, and carries target energy $g_j \asymp j^{-b}$. Every learner incurs an unseen-feature floor $\sum_j g_j (1 - p_j)^n \asymp n^{-(b-1)/s}$. An exactly solvable one-hot benchmark yields the three-frontier law

$$\mathcal{R}(m, n, K) \asymp m^{-(b-1)} + (\eta n K)^{-(b-1)/a} + n^{-(b-1)/s}.$$

Our main result extends this law to genuinely multi-active data, where replay updates do not commute. Example-norm-clipped SGD trains on every coactive row. Natural singleton rows provide coordinate-wise curvature, while clipping makes every rank-one update a contraction. The ordinary Polyak average attains the coverage rate up to a logarithm without a support-abort event; a label-free, distribution-aware anchor mask removes the logarithm and gives the sharp full frontier. In particular, all-data replay reaches the coverage floor after $O(\eta^{-1} n^{a/s-1})$ epochs. We also derive an input-only estimator of this useful replay horizon. Across 795 laptop-scale training runs, experiments expose all three frontiers, recover the predicted source exponents and scaling collapse, and predict the saturation epoch within a median factor below 1.7 across three spectral regimes.

Introduction

Power-law learning curves summarize how error changes with model size, unique data, and computation (Hestness et al. 2017; Kaplan et al. 2020; Hoffmann et al. 2022; Bahri et al. 2024). They are most consequential when one resource becomes scarce. In a data-constrained regime, additional compute must be spent either on processing new examples or on taking more optimization steps over examples already seen. Large-scale language-model experiments show that a few repeated epochs can initially resemble fresh data, but the marginal value of repetition eventually vanishes (Muenighoff et al. 2025). This creates a basic resource-allocation question:

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For a dataset of size n , how many epochs remain statistically useful?

A law expressed only through total sample updates nK cannot answer this question. Two runs can have the same update count while containing radically different amounts of unique information. The missing distinction is especially important for sparse or conditionally activated representations. A low-frequency feature may be hard to learn because its covariance is small, but it may also be completely absent from the fixed dataset. Additional epochs address the former problem and are powerless against the latter.

Existing linear-regression scaling theory characterizes how finite model width, sample size, and optimization time expose progressively weaker spectral directions (Lin et al. 2024; Lin, Wu, and Bartlett 2025; Chen and Zhou 2026). Multi-pass results show that reuse can improve the optimization term, while one-hot Zipfian models exhibit an eventual dataset-dependent plateau (Zou et al. 2022; Yan et al. 2026). Recent sparse-feature theory identifies unobserved coordinates as a distinct statistical bottleneck, but studies the solution after training to completion rather than the finite-epoch interaction between coverage and reuse (Sous and Winer 2026). Meanwhile, frequency-based accounts of neural scaling suggest that long-tailed skills or “quanta” are learned in decreasing order of occurrence (Michaud et al. 2023), and recent sparse-language-model experiments report that sparsity can shift the point at which repeated data saturate (Wu et al. 2026). These observations motivate a tractable theory in which frequency, optimization difficulty, and target importance are separately controllable.

We study sparse regression with three power-law sequences:

$$p_j \asymp j^{-s}, \quad \lambda_j \asymp j^{-a}, \quad g_j \asymp j^{-b}.$$

They respectively describe how often a feature appears, how quickly it can be optimized, and how much population error it contributes. The resulting learning process has three rank frontiers,

$$J_{\text{cap}} = m, \quad J_{\text{opt}} = (\eta n K)^{1/a}, \quad J_{\text{cov}} = n^{1/s}. \quad (1)$$

The smallest frontier determines the effective learned scale. This decomposition also yields an operational rule: repeat the data while J_{opt} is smallest; once it reaches $\min\{J_{\text{cap}}, J_{\text{cov}}\}$, another epoch cannot improve the leading error rate.

Our contributions are as follows.

- We prove an algorithm-independent Bayes and minimax unseen-feature lower bound, including its sharp power-law constant. We also derive a compact occupancy calculus for the number of observed features, the observed teacher norm, and the missing predictive energy.
- We derive an exact finite- n , finite- K one-hot replay law. It exposes the capacity, optimization, and coverage terms without an asymptotic surrogate for the SGD dynamics.
- For genuinely multi-active Bernoulli inputs, we resolve the noncommuting-update obstacle for a concrete all-row algorithm. Example-norm clipping gives pathwise contraction. Singleton rows that occur naturally in the data provide enough curvature to prove a sharp anchor-masked bound and a plain Polyak-average bound up to logarithms.
- We turn the theorem into an input-only stopping rule. Regressing conditional feature amplitude against empirical activation frequency estimates the exponent $a/s - 1$ and predicts K_{sat} before an expensive epoch sweep.
- We add direct tests of all three frontiers. In 795 training runs, the capacity sweep has slope -0.991 versus -1 , the coverage slope is -0.654 versus $-2/3$, a two-term law explains 98.1% of risk variation, and the input-only stopping estimate has median multiplicative error below 1.7 across three spectral settings.

The central message is simple but not captured by total compute: epochs improve what the dataset already contains; only unique examples move the coverage frontier.

Related Work

Empirical and mechanistic scaling laws. Early studies documented predictable power laws across language, vision, speech, and other modalities (Hestness et al. 2017; Kaplan et al. 2020; Rosenfeld et al. 2020). Compute-optimal scaling then made these curves into resource-allocation tools (Hoffmann et al. 2022). Theoretical accounts connect exponents to data-manifold dimension, kernel spectra, and task-model alignment (Sharma and Kaplan 2022; Bordelon, Canatar, and Pehlevan 2020; Canatar, Bordelon, and Pehlevan 2021; Bahri et al. 2024). The quantization model attributes power laws to a Zipf-distributed sequence of discrete skills learned in frequency order (Michaud et al. 2023), while the effective-frontier framework abstracts capacity, coverage, and optimization as competing truncations of a long tail (Zou et al. 2026). Our contribution gives these mechanisms a finite-data stochastic realization: it jointly analyzes unique occupancy and noncommuting replay dynamics, derives the replay horizon, and supplies an input-only estimator rather than assuming the frontier location.

Data constraints and repetition. Data-constrained language-model experiments show that modest repetition can be nearly neutral relative to fresh data, followed by sharply diminishing returns (Muennighoff et al. 2025). Recent work fits scaling laws that explicitly include repeated tokens, data mixtures, or sparse training; notably, sparse language models can postpone repetition-induced saturation (Wu et al. 2026). These studies establish the practical importance of the question but do not isolate which information repeated examples

cannot supply. Our contribution is a mechanism-level decomposition: nK controls an optimization frontier, whereas unique occupancy depends only on n .

Spectral regression and multi-pass SGD. Power-law linear regression provides sharp model-data-compute rates and a controlled setting for studying data reuse (Lin et al. 2024; Lin, Wu, and Bartlett 2025). Mini-batch and multi-pass extensions describe how sampling protocols and iterations alter the optimization rate (Chen and Zhou 2026; Zou et al. 2022). Zipfian one-hot regression produces a multi-epoch saturation law (Yan et al. 2026); we use that commuting model as an exact benchmark and then analyze multi-active inputs whose rank-one replay matrices do not commute. Our proof draws on iterate averaging and least-squares SGD (Polyak and Juditsky 1992; Bach and Moulines 2013; Defossez and Bach 2015; Dieuleveut and Bach 2016; Neu and Rosasco 2018; Ali, Dobriban, and Tibshirani 2020), but the central curvature argument is occupancy-driven rather than based on a uniformly well-conditioned empirical covariance.

Sparse features, occupancy, and missing mass. Sparse-feature scaling can differ qualitatively from dense spectral theory and can generate asymmetric interpolation phenomena (Sous and Winer 2026; Belkin et al. 2019; Hastie et al. 2022). Mathematically, our observed feature dictionary is an infinite-urn process (Karlin 1967). The unseen predictive term is a weighted missing mass, related to classical concentration results for unseen symbols and occupancy counts (Brend and Kontorovich 2013; Ben-Hamou, Boucheron, and Ohannessian 2017). We use regular variation not only to evaluate the missing mass, but also to couple it to the observed teacher norm that drives finite-time optimization.

Clipping and heavy-tailed optimization. Rare activations can create large conditional feature amplitudes even when the population covariance is trace class. Gradient clipping is widely used to control such heavy-tailed or locally unstable updates, and prior theory gives high-probability convergence guarantees under heavy-tailed noise (Gorbunov, Danilova, and Gasnikov 2020; Zhang et al. 2020). Our clipping rule has a more specific role: it enforces $\gamma_i \|X_i\|^2 \leq 1$, making every replay update a Euclidean contraction. This permits a transparent comparison between coactivation, singleton curvature, and feature occupancy.

Sparse Power-Law Regression

For $j \geq 1$, let

$$X_j = q_j Z_j E_j, \quad Z_j \sim \text{Bernoulli}(p_j), \quad E_j \sim \text{Rad}, \quad (2)$$

independently over coordinates and examples. Then $\lambda_j = \mathbb{E}X_j^2 = p_j q_j^2$. Labels are noiseless, $Y = \langle \theta, X \rangle$, and the predictive energy of coordinate j is $g_j = \lambda_j \theta_j^2$. We assume

$$p_j \asymp j^{-s}, \quad \lambda_j \asymp j^{-a}, \quad g_j \asymp j^{-b}, \quad (3)$$

$$s > 1, \quad a \geq s, \quad b > 1.$$

The condition $s > 1$ makes every example finitely supported almost surely. The condition $a \geq s$ makes $q_j^2 = \lambda_j/p_j \asymp$

$j^{-(a-s)}$ nonincreasing, which permits a rank-independent stable base step. Population excess risk is

$$\mathcal{R}(w) = \mathbb{E}_X[\langle w - \theta, X \rangle^2] = \sum_j \lambda_j (w_j - \theta_j)^2.$$

For an arbitrary measurable predictor, we use $\mathcal{R}(f) = \mathbb{E}_X[(f(X) - Y)^2]$.

Three frontiers and the correct intervention

Table 1 summarizes the regimes implied by Eq. (1). The same risk exponent $b - 1$ applies to each frontier because the unresolved predictive tail obeys $\sum_{j>J} g_j \asymp J^{1-b}$. What differs is the resource that moves J . This is why fitting a single law in the product nK can extrapolate correctly in the optimization regime and fail after coverage becomes active.

Phase boundaries and experimental design

The pairwise boundaries make the regime diagram directly testable. Capacity and coverage coincide when

$$m \asymp n^{1/s}; \quad (4)$$

capacity and optimization coincide when

$$\eta n K \asymp m^a; \quad (5)$$

and optimization reaches coverage when

$$K \asymp \eta^{-1} n^{a/s-1}. \quad (6)$$

These identities suggest three complementary sweeps. Varying m at large nK isolates the capacity exponent; varying K at fixed n reveals optimization followed by a plateau; varying n at large K measures the coverage exponent. A convincing empirical test should therefore change one frontier while keeping the other two safely larger. The experiments below follow exactly this design.

The exponent geometry also separates two notions that are easy to conflate. The target exponent b determines how much error remains beyond a learned rank, and hence changes every risk slope. By contrast, the epoch at which reuse saturates depends only on the input-side ratio a/s : a controls how weak the rare directions are, while s controls how quickly unique examples discover them. This is why the stopping horizon can be estimated from inputs without labels, even though the height of the final error floor remains task dependent.

Let D_N contain N unique examples and define the observed dictionary $\mathcal{S}_N = \cup_{i \leq N} \text{supp}(X_i)$, its size $V_N = |\mathcal{S}_N|$, the observed Euclidean teacher energy $A_N = \|\theta_{\mathcal{S}_N}\|_2^2$, and the missing predictive energy $U_N = \sum_{j \notin \mathcal{S}_N} g_j$. The following proposition makes explicit the three occupancy quantities used throughout the proof.

Proposition 1 (Occupancy calculus). *Assume exact profiles $p_j = c_p j^{-s}$, $\lambda_j = c_\lambda j^{-a}$, and $g_j = c_g j^{-b}$. Let*

$$\alpha = \frac{a-b+1}{s}, \quad \beta = \frac{b-1}{s}.$$

If $a - s + 1 < b < a + 1$, then $0 < \alpha < 1$ and

$$\mathbb{E}V_N \sim \Gamma(1 - 1/s)(c_p N)^{1/s}, \quad (7)$$

$$\mathbb{E}A_N \sim \frac{c_g}{c_\lambda} \frac{\Gamma(1-\alpha)}{a-b+1} (c_p N)^\alpha, \quad (8)$$

$$\mathbb{E}U_N \sim \frac{c_g}{s} \Gamma(\beta) (c_p N)^{-\beta}. \quad (9)$$

For two-sided power laws, the same quantities are respectively $\Theta(N^{1/s})$, $\Theta(N^\alpha)$, and $\Theta(N^{-\beta})$.

The identities follow by summing the inclusion probabilities $1 - (1 - p_j)^N$ or exclusion probabilities $(1 - p_j)^N$ and applying a rescaled Riemann-sum argument. The source interval has a concrete meaning. Its lower endpoint makes the observed Euclidean norm integrable at the coverage scale; its upper endpoint ensures that this norm actually grows with N . Moreover,

$$\alpha + \beta = \frac{a}{s}, \quad (10)$$

which is exactly the identity that later converts the finite-time optimization term into the saturation exponent $a/s - 1$.

For the information-theoretic lower bound, take independent Rademacher signs for the teacher coordinates while fixing their magnitudes.

Theorem 2 (Coverage floor). *Every measurable, possibly randomized learner trained on N unique examples satisfies*

$$\mathbb{E}_{\theta, D_N} \mathcal{R}(\hat{f}) \geq \sum_{j \geq 1} g_j (1 - p_j)^N \asymp N^{-(b-1)/s}.$$

For exact profiles, the leading constant is the one in Eq. (9). The same rate is a minimax lower bound over deterministic choices of the teacher signs.

Why the lower bound is algorithm independent. If coordinate j is absent from every training input, no observed label depends on its teacher sign. Conditional on the full dataset, all other teacher signs, and the learner's randomness, that sign remains symmetric. Averaging squared test error over the hidden signs leaves at least the unseen predictive energy. No amount of optimization, regularization, or replay can reduce this term.

Exact Replay Benchmark

In the one-hot benchmark, example i chooses $J_i \sim (p_j)$ and $X_i = E_i q_{J_i} e_{J_i}$. A width- m model is trained from zero for K passes with step η . Let $C_j \sim \text{Binomial}(n, p_j)$ count unique occurrences of coordinate j . The update matrices now commute, so the complete finite-sample risk is explicit.

Proposition 3 (Exact one-hot replay). *If $0 < \eta q_j^2 < 1$, then*

$$\mathbb{E}_{D_n} \mathcal{R} = \sum_{j \leq m} g_j [1 - p_j + p_j (1 - \eta q_j^2)^{2K}]^n + \sum_{j > m} g_j. \quad (11)$$

Under Eq. (3),

$$\mathbb{E} \mathcal{R} \asymp J_\star^{1-b}, \quad J_\star = \min\{m, (\eta n K)^{1/a}, n^{1/s}\}. \quad (12)$$

Equivalently, the risk is within constants of

$$m^{-(b-1)} + (\eta n K)^{-(b-1)/a} + n^{-(b-1)/s}.$$

Indeed, each occurrence multiplies the residual coefficient by $1 - \eta q_j^2$, and the binomial probability-generating function gives Eq. (11). The effective survival exponent is

$$np_j \{1 - (1 - \eta q_j^2)^{2K}\} \asymp \min\{np_j, \eta n K \lambda_j\}.$$

Table 1: The three resource frontiers. The active frontier is the smallest of the three, and therefore determines both the leading risk and the useful next intervention.

Active bottleneck	Rank scale	Leading risk	Resource that improves it
Capacity	$J_{\text{cap}} = m$	$m^{-(b-1)}$	increase model width
Optimization	$J_{\text{opt}} = (\eta n K)^{1/a}$	$(\eta n K)^{-(b-1)/a}$	take more stable updates
Coverage	$J_{\text{cov}} = n^{1/s}$	$n^{-(b-1)/s}$	collect unique examples

The first argument asks whether the feature appears; the second asks whether it has received enough optimization. Balancing the two at the coverage rank gives

$$K_{\text{sat}} \asymp \eta^{-1} n^{a/s-1}, \quad (13)$$

independent of the source exponent b . Proposition 3 is a sharp benchmark, but its diagonal updates avoid the main difficulty in multi-active data.

Coactive Replay SGD

For N selected examples, a sparse dictionary stores every identity in $\mathcal{S}_N$ whenever $V_N \leq m$. On the exponentially unlikely capacity-overflow event the algorithm returns zero; otherwise the selected problem is exactly realizable. At each update, sample a stored row uniformly with replacement and use

$$\gamma_i = \min \left\{ \eta, \frac{\rho}{\|X_i\|_2^2} \right\}, \quad (14)$$

$$w_{t+1} = w_t - \gamma_i X_i (X_i^\top w_t - Y_i).$$

where $0 < \rho \leq 1$, followed by $\bar{w}_T = T^{-1} \sum_{t < T} w_t$. We assume $\eta \sup_j q_j^2 \leq \rho$. Every sampled coactive row is used; singleton examples appear only in the analysis.

Let $C_{N,j}$ count examples whose support is exactly $\{j\}$. Because $\sum_j p_j < \infty$,

$$\begin{aligned} \pi_0 &= \prod_{k \geq 1} (1 - p_k) > 0, \\ \Pr\{\text{supp}(X) = \{j\}\} &= \frac{\pi_0 p_j}{1 - p_j} \asymp p_j. \end{aligned} \quad (15)$$

Fix $0 < \kappa \leq \pi_0/2$ and define the label-free, distribution-aware anchor set and output

$$\mathcal{A}_N = \{j : C_{N,j} \geq \kappa N p_j\}, \quad \hat{w}_T = P_{\mathcal{A}_N} \bar{w}_T. \quad (16)$$

The mask is applied only at readout and never filters a training row.

Theorem 4 (Sharp coactive replay and plain output). *Assume $a - s + 1 < b < a + 1$ and $N \leq c_m m^s$. Then example-norm-clipped, with-replacement replay satisfies*

$$\mathbb{E}\mathcal{R}(\hat{w}_T) \leq C \left[\frac{N^\alpha}{\eta T} + N^{-\beta} + e^{-cm} \right]. \quad (17)$$

The ordinary unmasked Polyak average, with no singleton-support abort or zero fallback other than capacity overflow, satisfies

$$\mathbb{E}\mathcal{R}(\bar{w}_T) \leq C \left[\frac{N^\alpha}{\eta T} + \{\log(eN)\}^{a/s} N^{-\beta} + e^{-cm} \right]. \quad (18)$$

The expectation is over the unique dataset and replay indices.

Proof anatomy

The proof separates stability, empirical curvature, and occupancy. Let $\delta_t = w_t - \theta_{\mathcal{S}_N}$. Since $\gamma_i \|X_i\|^2 \leq 1$, every rank-one update contracts pathwise:

$$\|\delta_{t+1}\|^2 \leq \|\delta_t\|^2 - \gamma_i (X_i^\top \delta_t)^2. \quad (19)$$

This is the point at which clipping resolves rare, high-norm coactive examples: they can slow an update but cannot increase the Euclidean error.

A singleton- j row is not clipped under the base-step assumption. Conditional on the fixed dataset, average Eq. (19) over the next replay index, retain only singleton contributions, and telescope. This gives the energy inequality

$$\frac{\eta}{N} \sum_{t < T} \mathbb{E}_{\text{sgd}} \sum_j C_{N,j} q_j^2 \delta_{t,j}^2 \leq A_N. \quad (20)$$

On $\mathcal{A}_N$, the coefficient satisfies $C_{N,j} q_j^2 / N \geq \kappa p_j q_j^2 = \kappa \lambda_j$. Jensen's inequality for the Polyak average therefore converts Eq. (20) into

$$\mathbb{E}_{\text{sgd}} \sum_{j \in \mathcal{A}_N} \lambda_j (\bar{w}_{T,j} - \theta_j)^2 \leq \frac{A_N}{\kappa \eta T}.$$

For the masked coordinates, a Chernoff bound gives $\Pr(j \notin \mathcal{A}_N) \leq e^{-cN p_j}$, hence

$$\mathbb{E} \sum_{j \notin \mathcal{A}_N} g_j \lesssim \sum_j j^{-b} e^{-cN j^{-s}} \asymp N^{-\beta}.$$

Finally, Proposition 1 supplies $\mathbb{E}A_N \asymp N^\alpha$, proving Eq. (17).

The plain output uses the same energy inequality on a simultaneous-coverage core $J_N \asymp (N/\log N)^{1/s}$. On the represented tail, pathwise contraction implies $\sum_{j > J_N} \lambda_j \delta_j^2 \leq \lambda_{J_N+1} A_N$. Taking expectations gives

$$\lambda_{J_N+1} \mathbb{E}A_N \asymp \{\log(eN)\}^{a/s} N^{-\beta},$$

where Eq. (10) is essential. A weighted occupancy-moment bound and Hölder's inequality control the rare datasets on which the core lacks singleton anchors. Thus no abort event is built into the plain algorithm. The supplement gives all constants, the moment lemma, and the capacity-overflow bound.

Corollary 5 (All-data saturation). *With $N = n$, $T = nK$, and $m \gtrsim n^{1/s}$,*

$$\mathbb{E}\mathcal{R}(\hat{w}_{nK}) \lesssim \frac{n^{\alpha-1}}{\eta K} + n^{-\beta}.$$

Consequently,

$$K_{\text{sat}} = O(\eta^{-1} n^{a/s-1}) \quad (21)$$

epochs suffice to reach the information-theoretic coverage rate. The plain average obeys the same power law up to the logarithm in Eq. (18).

Corollary 6 (Full frontier). *Given n, m, T , select*

$$N_{\star} \asymp \min\{n, m^s, (\eta T)^{s/a}\}$$

unique examples and run anchor-masked clipped replay. Then

$$\mathbb{E}\mathcal{R}(\hat{w}_T) = O(m^{-(b-1)} + (\eta T)^{-(b-1)/a} + n^{-(b-1)/s} + e^{-cm}). \quad (22)$$

The unmasked output attains the same exponents up to logarithms.

The progressive subset is a conservative achievability construction rather than a claim that discarding data is optimal. At fixed update compute, the experiments consistently favor more unique examples.

Budgeted Data Acquisition and Replay

The preceding corollaries treat unique examples and updates as independently specified resources. A more operational question is how to divide one budget between acquiring fresh examples and replaying the examples already available. Let one unique example cost $c_d > 0$, let one SGD update cost $c_u > 0$, and impose

$$c_d N + c_u T \leq B. \quad (23)$$

Corollary 7 (Budget allocation). *For a sufficiently small constant c_0 , choose*

$$N_B = \left\lfloor c_0 \min\left\{n, m^s, (\eta B/c_u)^{s/a}, B/c_d\right\} \right\rfloor, \quad (24)$$

$$T_B = \left\lfloor \frac{B - c_d N_B}{c_u} \right\rfloor.$$

Then anchor-masked clipped replay satisfies

$$\mathbb{E}\mathcal{R}(\hat{w}_{T_B}) = O(m^{-(b-1)} + n^{-(b-1)/s} + (\eta B/c_u)^{-(b-1)/a} + (B/c_d)^{-(b-1)/s} + e^{-cm}). \quad (25)$$

The unmasked output has the same power laws up to logarithms.

The derivation makes the resource tradeoff transparent. The constant c_0 ensures $T_B \asymp B/c_u$, and the choice of N_B balances the replay term $N^\alpha/(\eta T)$ against the coverage term $N^{-\beta}$, using $\alpha + \beta = a/s$. Thus, when n and m do not bind and $a > s$,

$$N_B \asymp (\eta B/c_u)^{s/a}, \quad K_B = T_B/N_B \asymp B^{1-s/a}. \quad (26)$$

Only a sublinear number of new examples is required as the budget grows; most additional budget is eventually spent replaying them until optimization meets coverage. When $a = s$, the two scales are both linear and fresh data must retain a constant budget share. The schedule is independent of b : target smoothness changes how much error lies beyond a learned rank, whereas a and s determine where the resource frontiers cross. Equation (25) is a frontier-matched guarantee for this benchmark, not a universal claim that the same cost split is minimax for arbitrary representations or noisy tasks.

An Input-Only Replay Stopping Rule

Corollary 5 implies that the useful number of epochs depends on a and s , but not on b . This is not merely an exponent-level observation. For exact profiles,

$$q_j^2 = \frac{\lambda_j}{p_j} = c_q p_j^{a/s-1}, \quad c_q = c_\lambda c_p^{-a/s}. \quad (27)$$

The coverage frontier consists of features with $p_j \asymp 1/n$. Such a feature appears only $O(1)$ times per epoch and needs $K \eta q_j^2 \asymp 1$ to be optimized. Therefore

$$K_{\text{sat}} \asymp \frac{1}{\eta q^2(p = 1/n)}. \quad (28)$$

This identity suggests a label-free plug-in estimator. For every feature that appears at least $r_{\min}$ times, estimate its activation frequency $\hat{p}_j$ and its conditional second moment $\hat{q}_j^2$. Fit

$$\log \hat{q}_j^2 = \hat{c} + \hat{r} \log \hat{p}_j$$

and predict

$$\hat{K}_{\text{sat}} = \eta^{-1} \exp(-\hat{c}) n^{\hat{r}}. \quad (29)$$

The theorem assumes known p_j only for the sharp output mask; the stopping estimator itself uses observed inputs alone. A full plug-in theorem for unknown activation probabilities is left open, but the experiment below tests the finite-sample calibration directly.

Experiments

Setup

We generate the independent Bernoulli design exactly in sparse CSR form by first sampling each coordinate occupancy and then assigning its active rows. Unless noted otherwise,

$$p_j = \frac{\mu}{\zeta(s)} j^{-s}, \quad \lambda_j = j^{-a}, \quad g_j = j^{-b},$$

with $(s, a, b) = (1.5, 2, 2)$, expected active features $\mu = 1$, $\eta = 0.2$, clipping radius $\rho = 0.9$, and ambient dimension $d = 4096$ or 8192 . Risk in the simulated coordinates is computed exactly from the coefficient error and we add the analytic tail $\zeta(b, d+1)$. The primary protocol is clipped with-replacement replay. All main sweeps use five seeds. The original release contains 670 runs; the capacity and stopping diagnostics add 125 runs, for 795 runs and 5,400 recorded risk checkpoints in total.

Optimization and coverage frontiers

Figure 1 shows power-law optimization followed by a floor that moves only with unique data. At $n = 8192$, the early averaged-risk slope is -0.459 , close to the prediction $-(b-1)/a = -0.5$. At $K = 512$, the slope across dataset sizes is -0.6548 , versus $-(b-1)/s = -0.6667$. Fitting all 60 primary averaged-risk points to

$$A(\eta n K)^{-(b-1)/a} + B n^{-(b-1)/s} \quad (30)$$

gives $R^2 = 0.981$ and median relative error 9.5%, without fitting either exponent. The normalized curves collapse

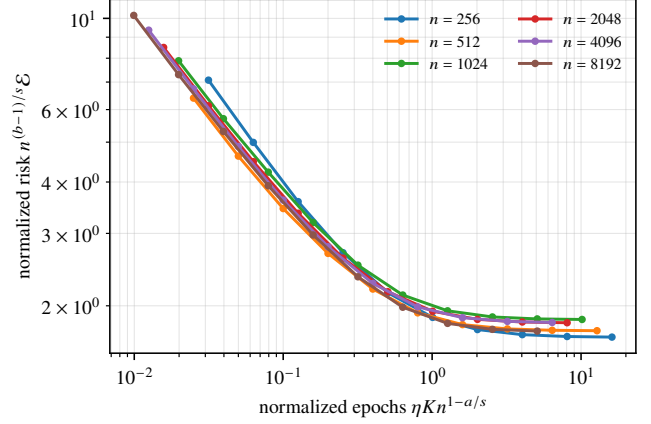
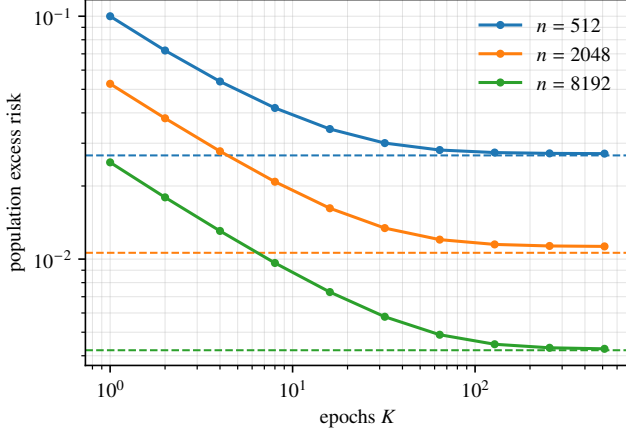


Figure 1: **Optimization gives way to coverage.** Left: averaged risk decreases with epochs and plateaus near the independently computed unseen-feature floor; increasing the number of unique examples lowers the plateau. Right: curves collapse after normalizing epochs by $\eta n^{1-a/s}$ and risk by $n^{(b-1)/s}$. Default $(s, a, b) = (1.5, 2, 2)$; means over five seeds.

around $\eta K n^{1-a/s} \asymp 1$, precisely the saturation scale in Eq. (21).

No experiment uses a singleton-support abort. The anchor mask is evaluated only as a conservative, label-free readout diagnostic; headline exponents and model comparisons use the ordinary unmasked output. In finite samples the plain output is consistently better because it retains features seen only in coactive rows, while Theorem 4 guarantees it up to a logarithm. Thus the mask is a sharp-proof device rather than an empirical performance trick.

Capacity frontier

The original experiments varied n and K but did not directly isolate J_{cap} . We therefore fix $n = 32768$, $K = 256$, and vary the representable rank $m \in \{64, 128, \dots, 4096\}$. For $m \leq 256$, averaged risk has slope -0.991 , while the exact omitted target tail has slope -0.996 ; both match the prediction $m^{-(b-1)} = m^{-1}$. Around $m \approx 1024$, the capacity tail falls below the unique-data floor and further width gives little benefit. The left panel of Figure 2 therefore exposes the third frontier that was implicit in the earlier scaling collapse.

Input-only stopping prediction

We use $r_{\min} = 8$ in Eq. (29). The measured stopping point is the first log-interpolated epoch at which the averaged risk is at most 1.25 times the exact unseen-feature floor. We test $(a, s, b) = (2, 1.5, 2)$, $(2.4, 1.5, 2)$, and $(2.5, 2, 2.5)$ over six dataset sizes. The mean fitted exponents $\hat{r}$ are respectively 0.349, 0.629, and 0.257, compared with the true values 0.333, 0.600, and 0.250. The median multiplicative errors in $\hat{K}_{\text{sat}}$ are 1.68, 1.49, and 1.55. These errors include both estimation noise and the finite-grid definition of “near the floor.” The right panel of Figure 2 shows that the estimator nevertheless tracks the change in replay horizon across data scales and spectral regimes.

Table 2: Predicted and measured coverage slopes.

b	predicted	measured
1.4	-0.267	-0.263
1.7	-0.467	-0.458
2.0	-0.667	-0.653
2.4	-0.933	-0.915

Additional checks

Progressive compute scaling. For total updates $T \in [2^{12}, 2^{20}]$, we choose $N = \lfloor (\eta T)^{s/a} \rfloor$ as in Corollary 6. The fitted slopes are -0.502 for the averaged iterate and -0.497 for the last iterate, matching $T^{-(b-1)/a} = T^{-1/2}$.

Source exponents. Holding $(a, s) = (2, 1.5)$ fixed, the large-epoch data slope tracks $-(b-1)/s$ for targets both rougher and smoother than the default (Table 2). The $b = 1.4$ case lies just outside the source range in Theorem 4 and is therefore a robustness test rather than a covered theorem setting.

Unique data at fixed compute. At $T = 2^{18}$ updates, increasing unique examples from 256 to 32768 lowers last-iterate risk by a factor of 12.8, even though total sample updates are unchanged. This directly rejects a law depending on nK alone after coverage becomes active.

Why clipping matters. Across expected coactivation $\mu \in [0.35, 2.35]$, clipped cyclic, reshuffled, and with-replacement replay approach the same floor. Under the aggressive setting $(\mu, \eta) = (2.25, 1.2)$, unclipped reshuffling reaches mean risk 1.16×10^{85} by four epochs and later overflows. Clipped replay remains stable and reaches 7.08×10^{-3} , near its 6.18×10^{-3} coverage floor. This stress test directly supports the pathwise mechanism in Eq. (19).

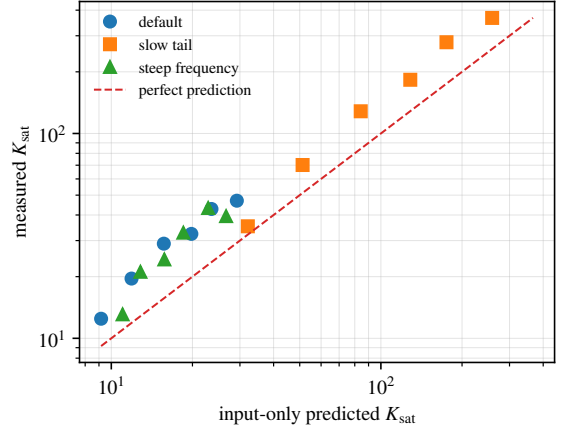
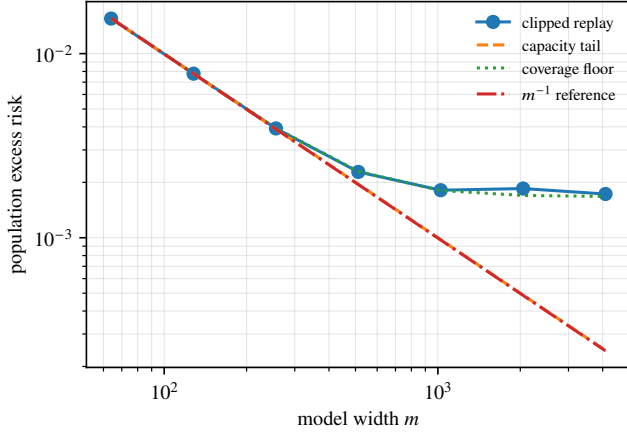


Figure 2: **The third frontier and an input-only stopping rule.** Left: at large n and K , risk follows the capacity tail m^{-1} before flattening at the unique-data floor; the fitted small-width slope is -0.991 . Right: the saturation epoch predicted from activation frequencies and conditional second moments tracks the measured crossing of 1.25 times the coverage floor across three spectral regimes. Each point aggregates five seeds.

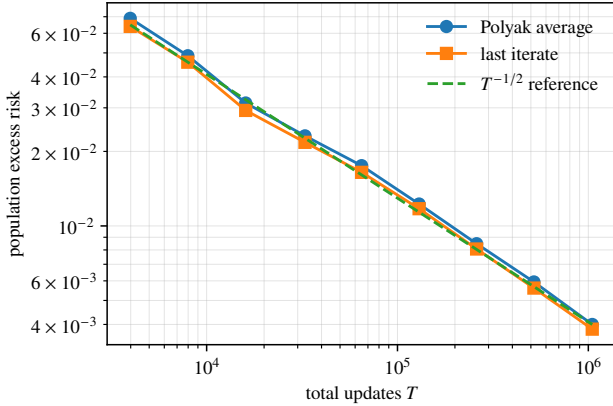


Figure 3: **Frontier-matched schedule.** Choosing $N \asymp (\eta T)^{s/a}$ yields the predicted $T^{-(b-1)/a} = T^{-1/2}$ compute law.

Local frontier diagnostic. Let $L = L_m + L_T + L_N$ denote the three terms in the additive law, with exponents $\delta = b - 1$, $\gamma = (b - 1)/a$, and $\beta = (b - 1)/s$. Their risk shares

$$\omega_m = \frac{L_m}{L}, \quad \omega_T = \frac{L_T}{L}, \quad \omega_N = \frac{L_N}{L}$$

satisfy $\omega_m + \omega_T + \omega_N = 1$. Equivalently, they are the normalized logarithmic elasticities $-\partial_{\log m} \log L/\delta$, $-\partial_{\log T} \log L/\gamma$, and $-\partial_{\log N} \log L/\beta$ (proof in the supplement). The largest weight identifies the active bottleneck; $\omega_T \asymp \omega_N$ recovers $K \asymp \eta^{-1} N^{a/s-1}$. Measured slopes probe all three corners: -0.991 versus -1 for width, -0.459 versus $-1/2$ for updates, and -0.655 versus $-2/3$ for unique data. Local finite differences therefore indicate which resource to expand next.

Discussion and Limitations

Interpretation of the stopping rule. The theorem predicts when optimization reaches the unique-data frontier, not the exact validation optimum in every noisy application. Label noise can create an earlier variance- or overfitting-limited optimum, so the coverage horizon is best viewed as an upper limit on useful unregularized reuse.

Known versus estimated frequencies. The sharp mask uses population p_j values; a plug-in mask needs uniform small-count control. The plain output needs neither a mask nor an abort event, at the cost of the logarithm in our present proof.

Representation and algorithmic scope. The theorem assumes independent addressable sparse coordinates, clipped with-replacement replay, averaging, noiseless labels, and $a - s + 1 < b < a + 1$. Fixed or learned features, cyclic replay, reshuffling, and last iterates require additional analysis. Progressive subsampling is an achievability construction, not an optimality claim.

Practical implication. Input occupancy and conditional second moments indicate when to redirect resources from replay to unique data or model capacity.

Conclusion

Sparse data impose a resource missing from total-update counts: unique feature coverage. We derive its unavoidable floor, characterize its interaction with capacity and optimization, and prove stable coactive replay reaches it at the predicted epoch scale. Width sweeps and an input-only estimator validate all three frontiers. Replay while optimization dominates; after coverage takes over, spend on unique data or model capacity.

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